

The Secret of Sainte-Chapelle

Thirteenth-Century Nanotechnology in Stained Glass

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Historical Context

- Built between 1243 and 1248 to house the **Passion relics** of King Louis IX.
- Contains 660 square metres of stained glass. The glazing took just six years.
- Original archives are lost. The physical material is the sole historical record.
- The recent restoration allowed **non-destructive** physical analysis on 110 original panels.

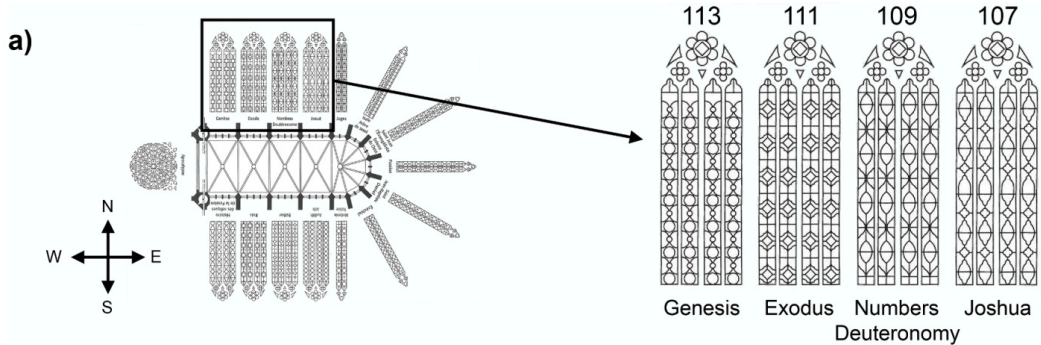


Figure 1: The spatial configuration of the four windows restored between 2011 and 2014 within the Sainte-Chapelle.

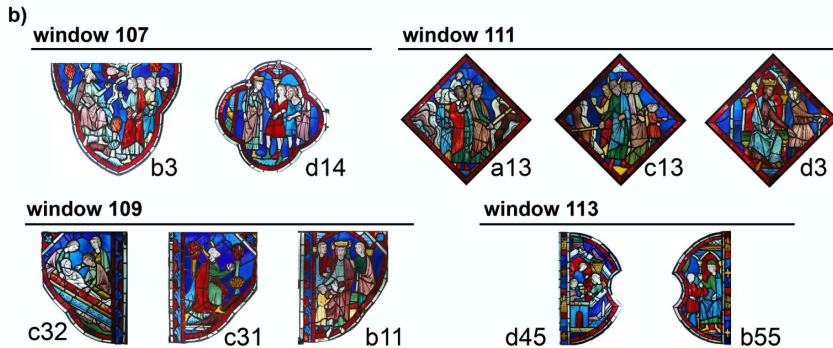


Figure 2: A selection of the historical glass panels analyzed.

Material Composition

The Glass Matrix and Network Modification

- The matrix is an amorphous solid. **Silica** acts as the primary network former.
- Plant ash provides potassium oxide. This acts as the **network modifier**.
- The glass has the highest silica fraction recorded in medieval Europe.
- This yields a rigid network. It requires high melting temperatures but ensures **exceptional durability**.

Compositional Topology

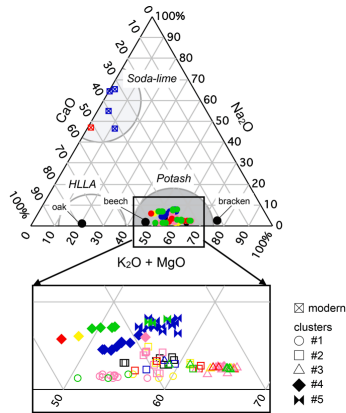


Figure 3: Ternary phase diagram illustrating the mass fractions of the primary glass constituents.

Optical Properties and Chromophores

Chromophores and Redox Equilibrium

- Macroscopic colour stems from transition metal ions. These act as localised **chromophores**.
- Colourless, yellow, and purple hues depend strictly on thermodynamic equilibrium.
- This relies on a delicate **redox balance** between iron (Fe^{2+} , Fe^{3+}) and manganese (Mn^{2+} , Mn^{3+}) states.

Optical Absorption in the Visible Spectrum

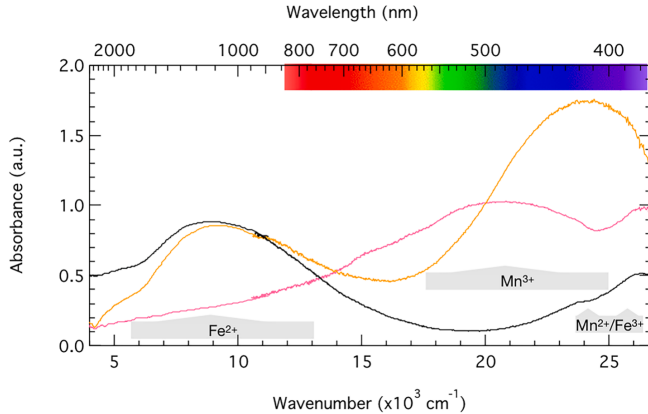


Figure 4: Optical absorption spectra for colourless, purple, and yellow glasses.

Transition Metal Concentrations

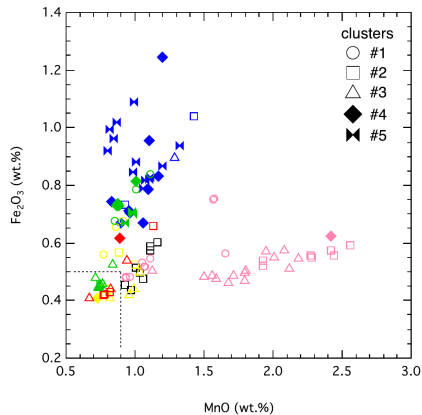


Figure 5: Concentration of iron versus manganese expressed as total oxides.

The Blue and Green Chromophores

- Blue transmission is dictated by Co^{2+} ions from imported German saffre.
- Green hue is a **superposition** of absorption states.
- High Cu^{2+} oxidises iron to Fe^{3+} . The copper and iron combine to define the green frequency.

Optical Absorption of Blue Glass

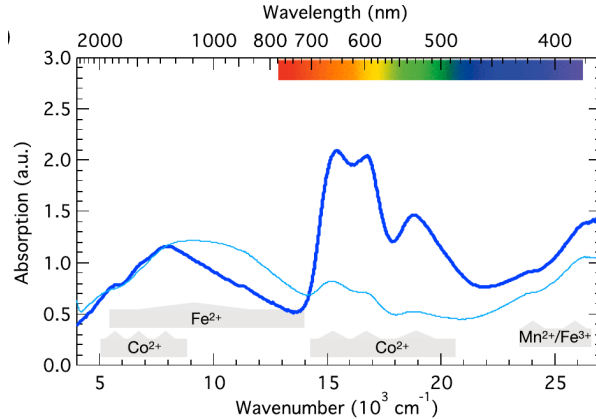


Figure 6: Optical absorption spectra contrasting light and dark blue glass matrices.

Trace Element Correlation in Blue Glass

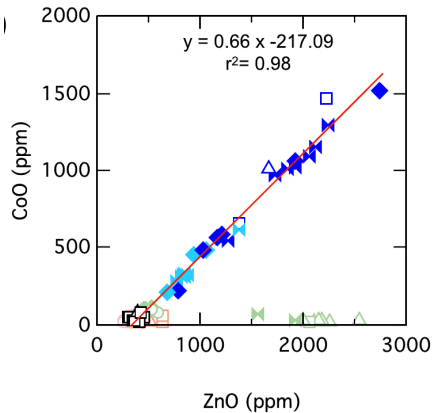


Figure 7: Concentration of cobalt relative to the amount of zinc oxide.

Optical Absorption of Green and Red Glasses

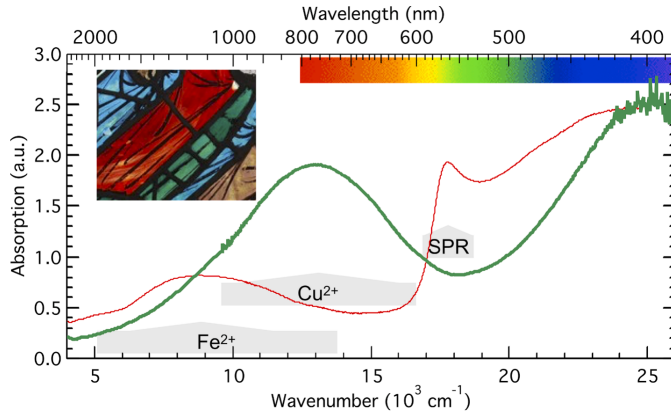


Figure 8: Optical absorption spectra for green and red glasses.

Trace Element Correlation for Copper Inclusions

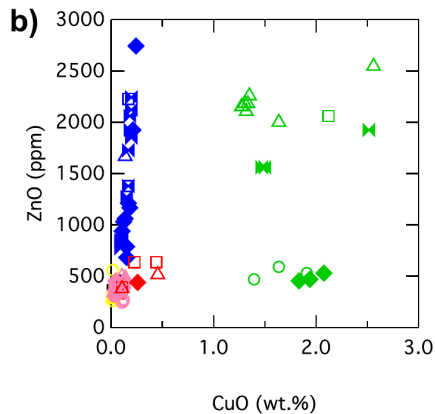


Figure 9: Amount of zinc oxide plotted against copper oxide.

Nanotechnology and Analytical Methods

- Red colour is not from dissolved ions. It originates from **metallic copper nanoparticles**.
- The electromagnetic field couples to the electron plasma, driving **surface plasmon resonance**.
- To maintain optical translucency, the glass is **striated**. It alternates red and colourless microscopic layers.

- The mean radius R of the copper inclusions is derived from the **full width at half maximum** (Γ) of the SPR band.
- Applying the Drude free-electron model to Mie scattering theory yields the boundary condition:

$$R = \frac{v_F}{\Gamma - \Gamma_0}$$

- Here, v_F is the **Fermi velocity** of the electron plasma in metallic copper.
- This calculation isolates the quantum confinement, determining the nanoparticles are approximately **15 nm** in size.

- Elemental characterisation used the AGLAE accelerator. This preserved the artefacts' topology.
- A 3 MeV proton beam induced inner-shell **ionisation** and nuclear **resonant transitions**.
- X-ray and gamma-ray emissions were recorded simultaneously to map elemental mass fractions.

- The proton beam traverses multiple layers. This yields an integrated, obscured signal.
- **Hierarchical cluster analysis** was applied to the emission data to isolate the base composition.
- Using Euclidean distances, this resolved **five distinct** compositional boundaries.

Conclusion

- The analysis successfully identified five clusters of **potash glass**.
- The glass matrix and chromophores show a uniform spatial distribution across all structural windows.
- This macroscopic uniformity proves the four windows were glazed **simultaneously**.
- It implies a single master atelier using a **unified material supply**.

References

- [1] Louis Grodecki and Catherine Brisac. *Le vitrail gothique au XIIIe siècle*. Fribourg, Suisse: Office du Livre, 1984. ISBN: 978-2-7191-0214-5.
- [2] Myrtille O.J.Y. Hunault et al. “Thirteenth-Century Stained Glass Windows of the Sainte-Chapelle in Paris: An Insight into Medieval Glazing Work Practices”. In: *Journal of Archaeological Science: Reports* 35 (Feb. 2021), p. 102753. ISSN: 2352409X. DOI: 10.1016/j.jasrep.2020.102753. (Visited on 19/02/2026).

Thank you.

Questions or comments?